FIE-Net: Foreground Instance Enhancement Network for Domain Adaptation Object Detection in Remote Sensing Imagery

Jun Zhang, Xupeng Zhang, Shuai Liu, Bin Pan, *Member, IEEE*, and Zhenwei Shi, *Senior Member, IEEE*

***Abstract*— Domain adaptation methods can mitigate performance degradation in remote sensing image object detection that arises from interdomain differences. However, current approaches often overlook the focused attention on foreground features, and the differences between the foreground and the background in remote sensing images implicitly diminish the characteristics of the foreground. To address this challenge, we propose a foreground instance enhancement network (FIE-Net) to balance the differences between the foreground and the background in remote sensing images, while enhancing the alignment and application of foreground features. The FIE-Net cooperates with the foreground-focused multigranularity feature alignment (FMA) module with the label filtering and application (LFA) module, progressively focusing on the salient foreground features during the processes of feature alignment and label application. Specifically, FMA directs feature alignment toward the foreground focus during the multigranularity feature alignment process, through foreground-focus perception attention and instance-centered emphasis approach. LFA balances the foreground and background difference information contained in labels, through the hybrid threshold label filtering method and the progressive label switching strategy. The experimental results indicate superior performance and generalization capabilities of our proposed FIE-Net in multiple remote sensing adaptation scenarios. The code is released at https://github.com/Lab-PANbin/**

***Index Terms*— Domain adaptation, foreground focus, object detection, remote sensing imagery.**

I. Introduction

NOWADAYS, object detection is a crucial aspect of remote sensing imagery interpretation, which involves locating and identifying objects of interest in remote sensing imagery. Remote sensing imagery object detection is one of the most fundamental tasks in civil and military intelligence systems, with widespread applications in natural resource management [1], intelligent agriculture [2], building detection [3], and terrestrial and naval traffic monitoring [4].

In recent years, several object detectors for remote sensing imagery [5], [6] have adopted complex two-stage R-CNN frameworks. There are several methods [7], [8], [9] that adapt to challenging remote sensing scenarios by incorporating external auxiliary branches or modules. Liu et al. [8] introduced scene context information related to the objects of interest in remote sensing and a scene auxiliary detection head for more accurate detection.

However, remote sensing imagery originates from a wide range of sources and exhibits significant interdomain variations, leading to suboptimal cross-domain detection performance. The diversity of acquisition conditions [10] for remote sensing imagery results in an inevitable discrepancy of data distribution between two datasets and disastrously restraints the improvement of model generalization further.

To mitigate the negative impacts of interdomain differences, several methods employ domain adaptation techniques to enhance the generalization capabilities of deep learning models. Domain adaptation for remote sensing imagery object detection can be categorized into metric-based [11], [12], adversarial-based [13], [14], [15], [16], [17], [18], [19], and generative-based [20], [21], [22], [23], [24], [25], [26]. The metric-based algorithms employ manually designed metric loss to minimize the discrepancy between feature distributions across domains. Koga et al. [12] designed the correlation alignment metric loss for a metric-based approach to remote sensing imagery. With the emergence of the gradient reversal layer (GRL) [27], adversarial-based algorithms employ a classifier as a domain discriminator, to align cross-domain distributions through adversarial training. Chen et al. [15] employed an adversarial-based approach to align the feature distributions of remote sensing imagery at both the prototype and image levels. Some generative-based algorithms employ GAN-based image transfer to generate target-style images from source-domain data for model training. Biswas and Tešic [20] employed CycleGAN to generate translated image data, assisting debiased contrastive loss function in eliminating biases in object detection. Additionally, some algorithms [28],

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[29], [30], [31], [32], [33] employ self-training methods to leverage the latent information present in the unlabeled target data. Self-training-based algorithms mostly employ a mean teacher framework, which comprises a teacher network and a student network. The teacher network predicts pseudo-labels for unlabeled target data, which serve as supervision for the student network. Zhu et al. [28] incorporate a dual-head co-training module into the mean teacher framework to reduce bias in pseudo-labels.

The differences between the foreground and the background in various remote sensing images affect the extraction of significant foreground features while existing methods often overlook this issue. The background features in remote sensing images are complex and often mix or obscure the foreground, while irrelevant background feature alignment implicitly diminishes instance characteristics. Furthermore, the variations in the foreground and the background differ among various types of remote sensing images. For instance, SAR images exhibit significant differences between the foreground and the background, while visible light images reveal smaller disparities. This results in challenges in effectively capturing instance features during cross-domain detection. Domain adaptation object detection in remote sensing imagery requires overcoming the challenges of foreground and background differences and enhancing the emphasis on instance characteristics.

In this article, we propose a foreground instance enhancement network (FIE-Net), a domain adaptation object detection for remote sensing imagery. The FIE-Net comprises two modules: the foreground-focused multigranularity feature alignment (FMA) module and the label filtering and application (LFA) module. The two modules, respectively, suppress background noise and emphasize foreground information in terms of features and labels. Specifically, FMA utilizes matrix reconstruction to mitigate the interference of redundant background features in instance-level feature alignment. Furthermore, deploying instance-focused modules on both pixel-level and instance-level feature alignment to reinforce the emphasis on foreground features. LFA employs a hybrid threshold for label filtering to eliminate labels that are primarily influenced by background noise. While utilizing label quality weighting during training to indirectly enhance the emphasis on labels that contain abundant instance features. Additionally, a progressive label-switching strategy enhances the effectiveness of instance feature knowledge transfer.

The main contributions of this article are as follows.

1) We propose the FIE-Net for domain adaptation object detection in remote sensing imagery, which enhances the emphasis on foreground features and balances the disparity between the background and the foreground.
2) We propose an FMA module, which effectively suppresses irrelevant background information while emphasizing significant foreground elements during the feature alignment process.
3) We design an LFA module that achieves a balance between the foreground and background information contained within the labels during the label utilization process.

The structure of the subsequent sections is as follows. In Section II, we present a comprehensive introduction to our FIE-Net. In Section III, we examine and analyze our experimental results. Finally, Section IV concludes this article.

II. Methods

The proposed FIE-Net is implemented in detail in this section, and the complete workflow of the FIE-Net is shown in Fig. 1.

A. Overview

1) Problem Definition: A labeled source domain $D_s = \{X_s, Y_s\}$ and an unlabeled target domain $D_t = \{X_s\}$ are provided for the algorithm. Y_s is the annotation of the source image X_s, which is composed of the bounding boxes B_s and corresponding categories C_s. The target image X_t is unlabeled, and its labels are unavailable. The cross-domain object detection technique is implemented utilizing D_s and D_t. The detector $\mathcal{M}$ can ultimately be obtained utilizing such an algorithm and perform properly in the target domain D_t.

2) Framework: As illustrated in Fig. 1, we proposed that the FIE-Net is supported by the mean teacher framework, where a teacher model is used to produce pseudo-labels to supervise the student model. The student model is trained with both the labeled source samples and pseudo-labeled target samples, where the labels of the pseudo-labeled target samples are generated by the teacher model. On the other hand, the teacher model is updated by using the exponential moving average (EMA) [34] of the weights of the student model.

In addition, we introduce the Fourier transform method to convert the source domain image to a style close to the target domain. The fast Fourier transform (FFT) is utilized to extract the amplitude and phase information from both the source domain and target domain images. By replacing the amplitude of the source domain image with the amplitude of the target domain image, a target-like domain image is generated. The phase of the image tends to preserve contour information, while the amplitude tends to preserve style information. The target-like domain image possesses a style that leans toward the target domain while retaining the contour details of the instances from the source domain. In this case, we can bridge the domain gap by such a target-like domain. Moreover, since target-like images inherit label annotations from the corresponding source images, they can be used as extra object supervision in the target-like domain. During the training process, the target-like domain participates in the training process in line with the target domain.

Usually, when training the mean teacher models, the labeled source samples are input into the student model after undergoing strong augmentation, and the detection loss is computed using the ground truth (GT) of the source domain. For the target samples, two different types of augmentation are used: weak augmentation and strong augmentation. The teacher model adopts weakly augmented images to generate reliable pseudo-labels, which are used to guide the learning of the student model with the input of strongly augmented images.

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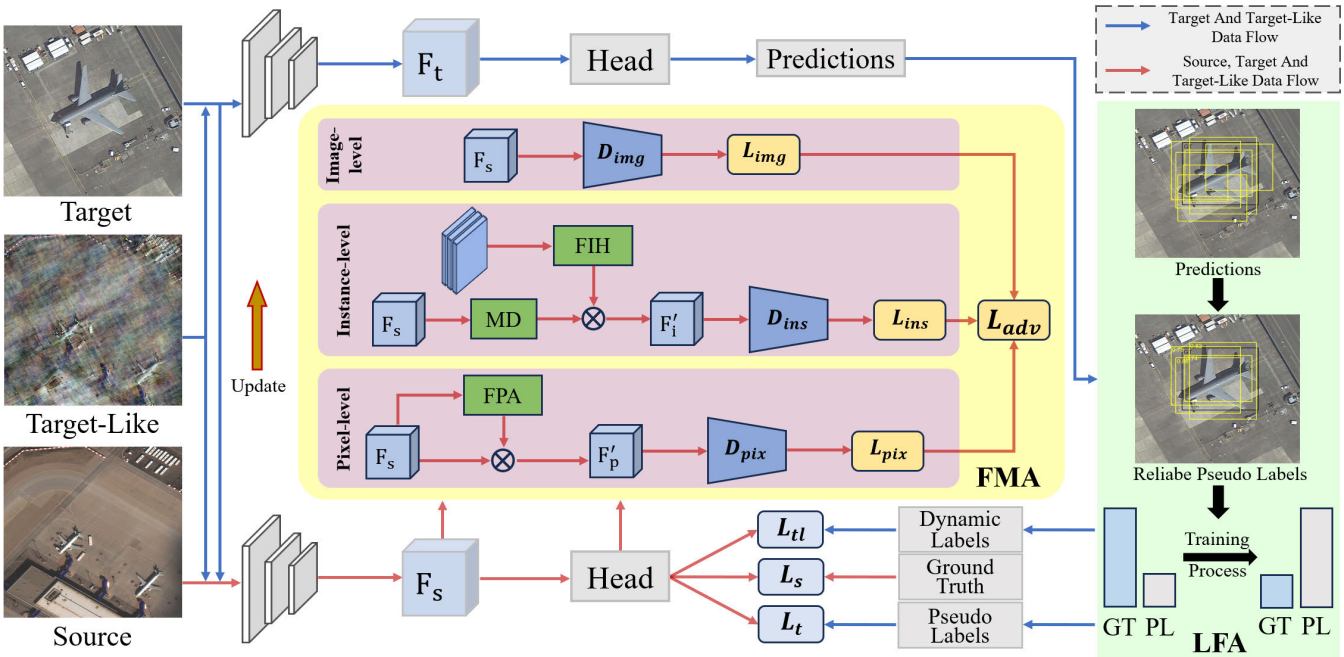


Fig. 1. FIE-Net is supported by the mean teacher framework and consists of a student model, which is the actual task model and a temporally ensembled version of the student model called the teacher model. The target-like domain is obtained from the source and target domains by the FFT method and is involved in the training along with the target domain. The student model is comprised of a base detector and the proposed FMA module. The teacher's model is updated by the EMA of the student's model and high-quality pseudo-labels are generated and applied using the newly proposed LFA module.

B. FMA Module

The FMA module consists of pixel-level, instance-level, and image-level feature alignment subnetworks, to distinguish the feature distribution between source and target domains in different perspectives, including pixel-level, instance-level, and image-level.

1) Pixel-Level Feature Alignment: Pixel-level adversarial learning, by focusing on each pixel, can better preserve the local information of the image, thereby assisting the model in identifying detailed domain-invariant features within the feature map. The domain discriminator, which learns at the pixel level, can classify each pixel based on its corresponding domain, thereby emphasizing the local features of the image. At the pixel-level feature alignment, pixel-level local information in object detection is not all transferable, such as the background. Forcefully aligning the untransferable information leads to negative transfer. Therefore, we introduce a foreground-focus pixel-level perception attention (FPA) module to focus on foreground transferable information and weaken the background untransferable information.

As shown in Fig. 2(a), given an input feature map F_s extracted from the backbone, we first apply average-pooling and max-pooling operations for downsampling and then concatenate them to generate an efficient feature descriptor. Then, the feature descriptor is combined with the downsampling results of F_s max-pooling operation using elemental addition, to enhance the retention of instance texture features. A 1×1 convolution and element-wise multiplication are used to achieve a foreground transferable map through learning spatial attention. The foreground transferable attention map F'_p is given by

$$F'_p = A(F_s) \otimes F_s \tag{1}$$

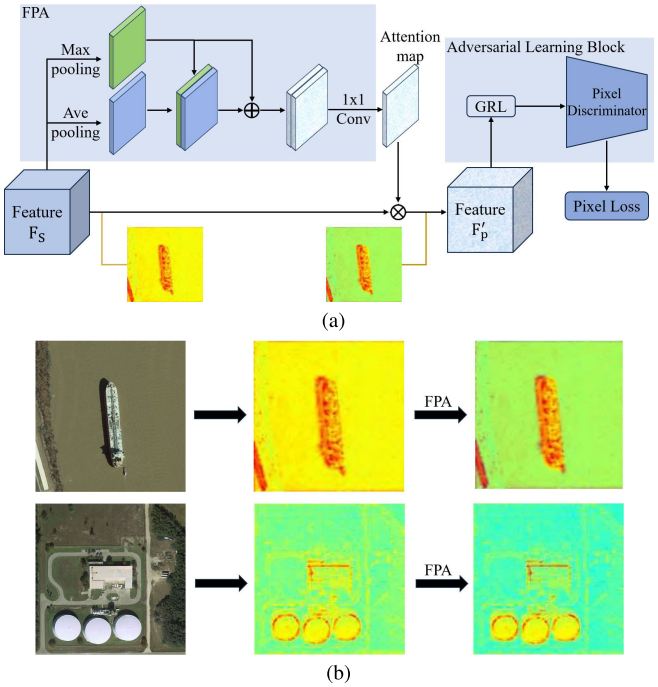


Fig. 2. (a) Pixel-level domain alignment subnet includes FPA and pixel-level adversarial learning block. (b) Visualization schematic of feature manipulation in pixel-level domain alignment subnet. In the visualization effect, orange indicates a higher amount of information embedded within the features, while blue indicates a lower amount of embedded information.

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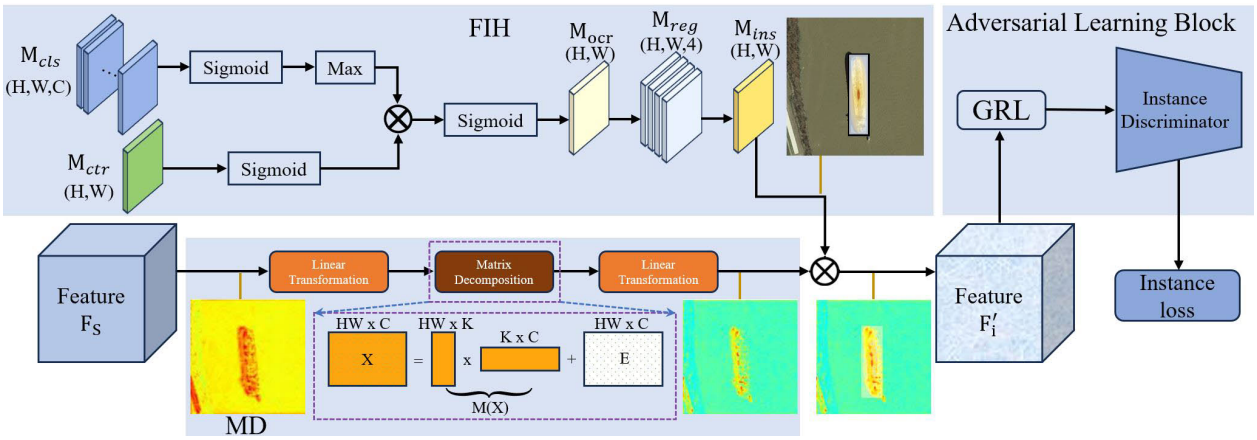


Fig. 3. Instance-level domain alignment subnet includes a reconstruction-based background feature reduction module (MD), an FIH module, and instance-level adversarial learning block.

where $A(\bullet)$ denotes the FPA and $\otimes$ denotes element-wise multiplication. The F_p' has a more significant difference between the foreground and the background, and negative transfer caused by the background could be addressed, resulting in more effective pixel-level feature alignment. As shown in Fig. 2(b), the background feature information is weakened to a certain degree after passing through the FPA module, resulting in a reduced ability to influence the domain discriminator. The foreground feature regions, which possess instance-specific details that are more worthy of preservation, have been well-preserved after passing through the FPA module. The FPA module enhances the distinction between foreground and background feature information, facilitating the domain discriminator in making accurate distinctions.

After achieving foreground transferable information, pixel-level adversarial learning is used to further align local domain-invariant features, through GRL and a pixel-level domain discriminator. The pixel-level domain discriminator is inclined to discern whether each pixel originates from the source domain or the target domain. The GRL reverses the backpropagation gradients from the discriminator and transmits them to the backbone layer responsible for feature extraction. In this way, the backbone layer strives to make the extracted features as similar as possible between the source and target domains, thereby capturing domain-invariant features. Specifically, a pixel domain discriminator D_{pix} tries to distinguish which domain the foreground transferable attention feature F_p' comes from, while the shallow layers of the backbone aim to confuse the discriminator. Formally, the loss function of pixel-level adversarial learning $\mathcal{L}_{\text{pix}}$ can be written as

$$\mathcal{L}_{\text{pix}} = \min_{\theta_{D_{\text{pix}}}} \max_{\theta_B} \frac{1}{nWH} \sum_{i=1}^{n} \sum_{w=1}^{W} \sum_{h=1}^{H} \times D_{\text{pix}}\left(F_{si}'\right)_{wh}^2 + \left(1 - D_{\text{pix}}\left(F_{ti}'\right)_{wh}^2\right) \tag{2}$$

where s and t, respectively, denote the source and target domains and n represents the number of input images. F_{ti}' and F_{si}' are the ith foreground transferable attention feature maps from the target and source domain, respectively. w and h are the coordinates on the above feature maps. During training, to obtain domain-invariant features, the network seeks the parameters θ_B of the shallow layers of the backbone via maximizing the loss, while simultaneously seeking the parameters $\theta_{D_{\text{pix}}}$ of the domain discriminator via minimizing the loss.

Deploying FPA in pixel-level adversarial learning, the pixel-level subnet concentrates on aligning domain-invariant features in the foreground regions, thereby improving attention to foreground instances in the detection task.

2) Instance-Level Feature Alignment: As shown in Fig. 3, two adaptive modules—the reconstruction-based background feature abatement (MD) module and the focus-centers for instances highlighting (FIH) module—are designed to explore and align instance-level foreground information. The instance-level feature alignment subnet mitigates the interference of background feature information and enhances the representation of instance feature information. Instance-level adversarial learning focuses on aligning feature information within instance bounding boxes, which can help the model identify domain-invariant features of the entire instance.

Due to the complex structure of remote sensing imagery, instance-level feature alignment is interfered with by complex background features. The MD module employed in this article reconstructs the features to suppress redundant background features. Specifically, inspired by Geng et al. [35] and Gao et al. [36], features are projected into a low-dimensional space by matrix reconstruction, thus eliminating the interference of the background on the features. For the feature mapping $F_s \in R^{(C\times H\times W)}$, extracted from the backbone, we first channel plane it to obtain F_f with shape $C \times HW$. F_f can be represented by the dictionary matrix $D_{\text{dict}} = [d_1, \ldots, d_k] \in R^{(C\times K)}$ to represent the corresponding code $C = [c_1, \ldots, c_{HW}] \in R^{(K\times HW)}$ in the following way:

$$F_f = F_f' + E = D_{\text{dict}} C + E \tag{3}$$

where F_f' is the low-rank reconstruction of F_f and E is the noise matrix we want to ignore in the alignment. As shown in Fig. 4, the reconstructed features significantly weaken the intensity of background information, achieving a good

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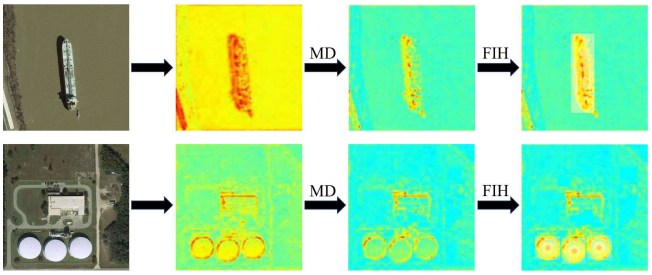


Fig. 4. Visualization schematic of feature manipulation in instance-level domain alignment subnet. In the visualization effect, orange indicates a higher amount of information embedded within the features, while blue indicates a lower amount of embedded information.

effect in reducing background noise. However, the foreground information is also affected after reconstruction, with some feature information not being well-preserved.

The reconstructed features need to be integrated with the FIH module to emphasize instance domain-invariant features. The FIH module is designed to facilitate the identification of instance center points and assist in predicting bounding box distance parameters. Furthermore, important background information will undergo a certain degree of attenuation after reconstruction. Background information can be classified into significant background information that is somewhat beneficial to detection and background noise information that interferes with detection. Important background information is often distributed around foreground instances and typically located within the instance bounding box. The FIH module is capable of enhancing the representation of background features within the bounding box, further attenuating background noise information outside the bounding box, and preserving significant background features.

As shown in Fig. 3, to find the confident area containing foreground objects, we apply three cues obtained in the detection header $M_{\text{cls}} \in R^{H\times W\times C}$, $M_{\text{ctr}} \in R^{H\times W}$, and $M_{\text{reg}} \in R^{H\times W\times 4}$, so that the alignment can focus more on object parts. First, we apply sigmoid activation to M_{cls} and M_{ctr} on each channel, perform a max operation on the category of M_{cls}, and multiply the two and then apply sigmoid activation again to obtain the object-centered region M_{ocr}. Mapping the high probability central region of M_{ocr} onto the distance map M_{reg} yields the corresponding bounding box. We assign a value to each point in the bounding box, the further away from the center the lower the value, to obtain the instance-aware map M_{ins} of the focus center. Using the instance-aware map centered on focus, M_{ins}, we can highlight the instance regions on the feature map that should be aligned with particular attention. As shown in Fig. 4, the FIH module enhances the foreground information that is not well-preserved after reconstruction, thereby improving the damaged instance feature information. Additionally, the focus-on-center approach assigns the highest weight to the center of the instance, assisting the model in determining the center of the bounding box. The holistic weighting within the bounding box not only strengthens the overall features of the instance but also affects the significant background information surrounding the instance, aiding in the localization of the bounding box.

The instance-aware map M_{ins} focused on the center and the reconstructed feature map F'_f are combined using element-wise multiplication, to obtain the feature map F'_i. The feature map F'_i is input into the instance-level adversarial learning block, where it further aligns instance-domain invariant features via a GRL and instance-level domain discriminator. Formally, the loss function for instance-level adversarial learning $\mathcal{L}_{\text{ins}}$ can be written as

$$\mathcal{L}_{\text{ins}} = \min_{\theta_{D_{\text{ins}}}} \max_{\theta_B} \frac{1}{n} \sum_{i=1}^{n} \log D_{\text{ins}}\left(M^s_{\text{ins}} \otimes F'_{si}\right) + \log\left(1 - D_{\text{ins}}\left(M^t_{\text{ins}} \otimes F'_{ti}\right)\right) \tag{4}$$

where s and t denote the source and target domains, respectively, and n denotes the number of input images. F'_{si} and F'_{ti} are the low-rank reconstructed feature maps of the source and target domains, respectively, and M^s_{ins} and M^t_{ins} are the center-of-focus instance-aware maps of the source and target domains, respectively.

3) Image-Level Feature Alignment: The goal of image-level feature alignment is to align feature mappings at the image level to reduce domain gaps. Image-level adversarial learning can overcome the influence of image styles and reduce the differences in image-level features across different domains. We apply an image-level feature alignment subnet composed of a GRL and image-level domain discriminator D_{img}, after the backbone. Formally, the loss function for image-level adversarial learning $\mathcal{L}_{\text{img}}$ can be written as

$$\mathcal{L}_{\text{pix}} = -\min_{\theta_{D_{\text{pix}}}} \max_{\theta_B} \frac{1}{n} \sum_{i=1}^{n} \log\left(D_{\text{img}}(F_{si})\right) + \log\left(1 - D_{\text{img}}(F_{ti})\right) \tag{5}$$

where θ_B denotes the parameters of the whole backbone. F_{ti} and F_{si} are the ith output feature maps of the backbone from the target and source domains, respectively.

C. LFA Module

The self-training framework of Mean Teacher [37] generates pseudo-labels through the teacher network for the student network to learn. The balance between the differences in the foreground and the background of pseudo-labels has a certain impact on instance feature bounding box regression and object classification. Our proposed LFA module consists of a hybrid threshold for label filtering module and a progressive label switching strategy, which improves the LFA system for remote sensing imagery participating in the Mean Teacher self-training.

1) Hybrid Threshold for Label Filtering Module: We apply a hybrid threshold for the label filtering module to the pseudo-labels generated by the teacher model. Incorporating the pseudo-label quality assessment method of the Harmonious Teacher [38] network as a soft threshold. And introduce a dynamically adjusted threshold, based on the consistency of classification scores and localization scores for different categories as a hard threshold, thereby forming a label-filtering module with a hybrid threshold. This approach aims to maximize the utilization of hard examples while avoiding excessive

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attention on pseudo-labels of lower quality, ultimately enhancing the balance between the differences in foreground and background of pseudo-labels. The quality of the pseudo-labels is evaluated based on the harmony between classification scores and localization scores. This harmony is defined as the weighted result of the pseudo-label's classification score and localization score, which is formulated as follows:

$$h = p^{\beta} u^{(1-\beta)} \tag{6}$$

where p denotes the classification score and u is the localization score. β is a tradeoff parameter that balances the contribution between prediction score and localization score.

During the training process, a threshold is autonomously established and dynamically adjusted for the harmony degree. This approach preserves the advantages of weighted training with the harmony degree, fully utilizing the merits of predicted instances, while eliminating lower-quality predictions to prevent the network from paying excessive attention to biased predictions. Our approach enables the fine-tuning of thresholds for each class, maintaining stable changes that are consistent with the model's functionality, and ensuring the correct filtering of lower-quality pseudo-labels. We initialize the threshold τ_t as 0.5, for each training iteration, calculating the average harmony score $M_{h_t}(c)$ of all detections for a batch of unlabeled data. To update the threshold, we employ an EMA method based on $M_{h_t}(c)$. The threshold τ_t is defined and adjusted as follows:

$$\tau_t(c) = \begin{cases} 0.5, & \text{if } t = 0 \\ \lambda \tau_{t-1}(c) + (1-\lambda)\dfrac{1}{B}\displaystyle\sum_{b=1}^{B} M_{h_t}(c), & \text{if otherwise} \end{cases} \tag{7}$$

where $\tau_t(c)$ is the threshold for class c, $\lambda \in (0, 1)$ is the momentum decay for EMA, and B represents the batch size of unlabeled data. The detailed process of pseudo-label filtering is provided in Algorithm 1. The harmony h for each pseudo-label is obtained through the classification score and localization score. The average harmony score $M_{h_t}(c)$ is calculated for each category, and, subsequently, the threshold $\tau_t(c)$ is updated using the EMA method. A pseudo-label is discarded if its coordination degree h falls below its category-specific threshold $\tau_t(c)$. The impact of the retained pseudo-labels on the loss is weighted by their respective coordination degrees h.

2) Progressive Label Switching Strategy: We have observed some limitations in training solely in the target-like domain using pseudo-labels, as the differences in the foreground and the background across domains hinder the student model's ability to learn effectively. Furthermore, training solely with the GT labels from the source domain fails to utilize the synthetic knowledge generated by the teacher model, thereby making it challenging to fully exploit the guiding role of label information in instance bounding box regression and target classification. To address these issues, we integrate the pseudo-labels generated by the teacher model with the GT labels into the training of the target-like domain. The approach is defined as follows:

$$\mathcal{L}_{\text{tl}} = \mu \mathcal{L}_{\text{GT}} + \lambda \mathcal{L}_{\text{PL}} \tag{8}$$

Algorithm 1 Train of the Hybrid Threshold for the Label Filtering Module. The Tuple (p, u) Represents the Set of Pseudo-Label Classification Scores and Localization Scores, h_c Encompasses the h Values for Each Category, and l_h Denotes the Loss for Pseudo-Labels With a Coordination Degree of h

for i=1; i ≤ epoch num; i++ **do**
  if i==1 **do**
    $\tau_t(c) = 0.5$;
  end if
  for p_c, u_c in (p, u) **do**
    $h_c = p_c^{\beta} u_c^{(1-\beta)}$;
  end for
  for h in h_c **do**
    $M_{h_t}(c) = \frac{1}{B}\sum_{b=1}^{B} h$;
  end for
  for M_{h_t} in $M_{h_t}(c)$ **do**
    $\tau_t(c) = \text{Update}(\tau_t(c), M_{h_t})$;
  end for
  for h in h_c **do**
    if $h \le \tau_t(c)$ **do**
      Exclude h;
    end if
  end for
  for h in h_c **do**
    $l_h = l_h * h$;
  end for
function Update($\tau_t(c)$, M_{h_t})
  $\tau_t(c) = \lambda \tau_{t-1}(c) + (1-\lambda)\frac{1}{B}\sum_{b=1}^{B} M_{h_t}(c)$;
  return $\tau_t(c)$;
end function

where μ and λ are hyperparameters that control the degree of utilization of GT labels and pseudo-labels during the training of the target-like domain. At the beginning of training, μ is set to 0.7 and λ is set to 0.3. During the training process, the value of μ is gradually decreased while the value of λ is increased. Specifically, after a certain number of training iterations, μ is reduced by 0.1 and λ is increased by 0.1. The weight adjustment stops when half of the training process is reached, at which point μ is 0.3 and λ is 0.7.

For the target-like domain data, which consists of both GT labels and pseudo-labels generated by the teacher model. At the beginning of the training, we set a higher weight for the GT labels and assigned a lower weight to the pseudo-labels, emphasizing the use of the correct GT labels during initial training. As the training progresses, we gradually decrease the weight of the GT labels while increasing the weight of the pseudo-labels. Ultimately, the focus shifts to training with pseudo-labels to leverage the knowledge from the teacher model. This combination makes the transfer of knowledge from the teacher model to the student model more effective. Our progressive label-switching strategy enables the target-like domain to serve as a better bridge between the source and target domains, initially resembling the training process of the source domain and gradually transitioning toward the training process of the target domain.

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TABLE I
DATASET OVERVIEW FOR DIFFERENT DOMAIN ADAPTATION SCENARIOS

domain adaptation scenarios	dataset	sensor	GSD	image size	images
between different optical satellites	DOTA	GoogleEarth+	0.1 ∼ 4.5	800 × 800 ∼ 4k × 4k	2806
	xView	WorldView-3	0.3	2.5k × 2.5k ∼ 5k × 3k	216
	DIOR	GoogleEarth	0.5 ∼ 3	800 × 800	16362
between satellite and UAV	UCAS-AOD	GoogleEarth	—	1280 × 659	510
	CARPK	UAV	0.01 ∼ 0.04	1280 × 720	1448
from optical to SAR images	HRRSD	GoogleEarth	0.15 ∼ 1.2	1k × 1k ∼ 2k × 2k	2165
	SSDD	RadarSat-2+	1 ∼ 15	300 × 300 ∼ 500 × 500	1160

D. Overall Optimization Objective

With the consideration of the losses in FMA as well as LFA, we summarize the total loss and formulate it as follows. Concerning the adversarial learning loss in the feature extraction phase, we summarize the adversarial loss as follows:

$$\mathcal{L}_{\text{adv}} = \alpha \mathcal{L}_{\text{pix}} + \alpha_1 \mathcal{L}_{\text{ins}} + \alpha_2 \mathcal{L}_{\text{img}}. \tag{9}$$

Besides, we also summarize the detection loss, which is composed of supervised loss and unsupervised loss and is formulated as follows:

$$\mathcal{L}_{\text{det}} = \mathcal{L}_s + \hat{L}_t + \mathcal{L}_{\text{tl}}. \tag{10}$$

The overall optimization objective of our FIE-Net can be written as follows:

$$\mathcal{L} = \mathcal{L}_{\text{adv}} + \eta \mathcal{L}_{\text{det}}. \tag{11}$$

III. EXPERIMENTS

A. Datasets

In this study, we evaluated the effectiveness and generalization capabilities of the FIE-Net under three widely adopted domain adaptation scenarios by dividing seven public remote sensing datasets into three groups. The source and target datasets used in our designed scenarios are open access. First, we evaluated the cross-domain knowledge transfer of a large number of categories between different satellites for the detector (DIOR [39] → DOTA-10 [40]). Then, we evaluated the cross-domain knowledge transfer of a small number of categories by the detector between different satellites (xView [41] → DOTA-3 [40]). To validate the generalization ability of our approach, we demonstrate the effectiveness of the proposed between satellites and UAV (UCAS-AOD [42] → CARPK [43] and CARPK → UCAS-AOD). Furthermore, we have followed and evaluated the performance of our method in a cross-modalities domain adaptation scenario, transitioning from optical images to SAR images (HRRSD [44] → SSDD [45]). These datasets are compared in Table I and presented in the following paragraphs.

1) DOTA: The DOTA dataset, published by Wuhan University, includes 2806 aerial images from various sensors and platforms (e.g., Google Earth). A total of 188 282 instances were annotated by experts, and forms covering 15 common object classes such as airplanes, ships, and vehicles. We crop the original images into 800 × 800 pixels with an overlapping step of 100 pixels as input to the network. To be consistent with the xView remote sensing dataset in terms of categories, we selected three categories for the experiments in DOTA: airplane, ship, and storage-tank to form the DOTA-3 dataset; similarly, we selected ten categories: airplane, ship and storage-tank, and so on, forming the DOTA-10 dataset.

2) xView: This dataset is used alongside DOTA-3 to evaluate the method's knowledge transfer in a small number of categories between different satellites. The xView dataset was captured by the WorldView-3 satellite and contains more than one million annotated instances. To be consistent with DOTA's categories, we have merged the fine-grained tags. For example, motorboats, sailboats, and fishing boats were merged into the ships category. Our experiments use an officially segmented training set and focus on the airplane, ship, and storage tank categories. We crop the original image to 800 × 800 pixels as input to the network. The cropped xView dataset has a total of 3287 images and 26 807 instances.

3) DIOR: The DIOR dataset contains 23 463 optical remote sensing imagery and 273 248 objects with 20 common object classes. The DIOR dataset has variations for each object class in terms of viewpoint, occlusion, appearance, background, and object pose. We extracted the same ten classes with 16 362 images and 225 965 objects from the DIOR dataset as in the DOTA-10 dataset.

4) UCAS-AOD: The dataset was released by the University of Chinese Academy of Sciences (UCAS) in 2014 and acquired from Google Earth. We focus on the car data, consisting of 510 images and 7114 vehicles, and the image size is 1280 × 659 pixels. In addition, the dataset was not divided uniformly by official sources. To ensure the reproducibility of the results, 70% of the images were randomly selected as the test set and the remaining images were used for training.

5) CARPK: The public CARPK dataset is captured by the unmanned aerial vehicle (UAV) in parking lots, consisting of 1448 images and 89 774 vehicles. Each image is 1280 × 720 pixels in size and contains only a single class of vehicles.

6) HRRSD: The high-resolution remote sensing detection (HRRSD) dataset was released by the University of Chinese Academy of Sciences in 2019. For our experiments, we chose the ship data, which includes 2165 images and contains a total of 3975 ships. We adjust the minimum Short Edge Length Of The Input Image To A Maximum of 800 pixels.

7) SSDD: The SAR ship detection dataset (SSDD) is a publicly available dataset created by Li et al. [45]. It contains a total of 1160 images, including 2456 ships. In our experiments, we apply the original image size as input.

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B. Evaluation Metrics

In this article, we employ standard evaluation metrics for detection tasks to evaluate the performance of models. Specifically, average precision (AP) is utilized, which represents the area under the precision–recall curve and combines precision and recall. The expression is as follows:

$$\mathrm{AP} = \int_0^1 P(R) dR \tag{12}$$

where P represents the precision rate and R is the recall rate. We ultimately used mean AP (mAP) to evaluate the performance of the detector, where mAP represents the average AP value of multiple categories, as follows:

$$\mathrm{mAP} = \frac{1}{c} \sum_{i=0}^{c} \mathrm{AP}_i \tag{13}$$

where c represents the number of categories.

C. Implementation Details

Following, we take FCOS [46] with the IoU branch as the base detector for experiments. The VGG16 [47] pretrained on ImageNet [48] is adopted as the backbone. We use batch-size 8 for both the source and target images and 1 RTX 3090 GPU. The learning rate is set to 0.005 without any decay. We pretrain the model 10 k iterations with labeled source samples to initialize the teacher and student models and continuously train the model 30 k iterations for both the source and target domain. The shorter side of the images is resized at most 800 followings. For data augmentation, we applied random horizontal flips to weak enhancements and randomly added color dithering, greyscale, Gaussian blur, and cut-out patches to strong enhancements. The smoothing hyperparameter of the EMA [34] is set to 0.999. The hyperparameters β in (6) is set to 0.5. The weights η in (11) are set to 1.0. The values of the hyperparameters were determined through experimentation, with specific ablation studies outlined in Tables II and III.

D. Ablation Studies

In this section, to effectively analyze the contribution of each component in our proposed FIE-Net, we conducted ablation experiments by replacing or removing parts of the components in xView for DOTA task adaptation scenarios.

1) Effectiveness of Each Component: To verify the effectiveness of each component in FIE-Net, we equipped individual modules of FMA and LFA, respectively. We first examine the contribution of FMA. Next, we describe the advantages of using LFA. In addition, we explore the benefits of self-training using LFA for target-like domain data and demonstrate the effectiveness of the proposed LFA. The experimental results of the ablation experiments are given in Table IV.

Training HT [38] directly on remote sensing imagery is challenging in obtaining favorable results due to the complex foreground and background features of remote sensing imagery. Compared with the baseline network, our proposed FMA cuts down the background feature interference and strengthens the foreground feature representation, which is more in line with the requirements of remote sensing imagery feature alignment. It can show better cross-domain alignment at the feature level, achieving a 5.5% improvement in mAP. Then, we apply the LFA module alone on the baseline network and improve the overall performance by 3.2%. The combination of the FMA and LFA modules resulted in a performance improvement of 7%, achieving an accuracy of 52.4%. The experimental results demonstrate that the individual application or combined use of the FMA module and the LFA module can both achieve improvements in detection outcomes. The results above clearly indicate the effectiveness of both the FMA and LFA modules.

In addition, we explored the effectiveness of using the LFA for self-training on target-like domain data. Porting the target-like domain to HT [38] resulted in a 4.3% improvement in mAP. Selecting valuable labels for weighted self-training through the LFA module, mixing pseudo-labels and GT labels of the class-target domain, and assigning gradual weights with the training process can further improve the mAP by 1.5%. The use of FFT to generate synthetic target domains and the FMA module during training achieved a performance of 53.7%. Building on this, the introduction of the LFA module resulted in an improvement of 1.4% in mAP, ultimately reaching a performance of 55.1%. Compared to the baseline, the overall performance increased by 9.7%. The experimental results indicate that, based on the target-like domain generated

TABLE II
ABLATION EXPERIMENTS OF HYPERPARAMETERS β IN DIFFERENT DOMAIN ADAPTATION SCENARIOS

β	0	0.25	0.5	0.75	1.0
xView→DOTA-3	53.9	54.6	**55.1**	54.3	53.8
UCAS-AOD→CARPK	77.5	78.1	**79.4**	78.7	77.6
HRRSD→SSDD	57.0	57.8	**58.5**	58.2	56.9

TABLE III
ABLATION EXPERIMENTS OF HYPERPARAMETERS η IN DIFFERENT DOMAIN ADAPTATION SCENARIOS

η	0.8	0.9	1.0	1.1	1.2
xView→DOTA-3	53.5	54.9	**55.1**	53.7	53.1
UCAS-AOD→CARPK	75.6	77.5	**79.4**	78.8	77.1
HRRSD→SSDD	55.7	57.0	**58.5**	58.4	57.2

TABLE IV
ABLATION STUDY OF OUR PROPOSED FIE-NET

Method	FMA	FFT	LFA	mAP_{50}
baseline				45.4
Proposed	✓			50.9
		✓		49.7
			✓	48.6
	✓	✓		53.7
	✓		✓	52.4
		✓	✓	51.2
Ours	✓	✓	✓	**55.1**

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TABLE V
ABLATION STUDY OF OUR PROPOSED LFA

Method	GT	PL	Mixed Labels	Dynamic Labels& Gradual Weight
FIE-Net	53.9	53.3	53.4	**55.1**
LFA	47.4	45.9	46.7	**48.6**

by FFA, the individual application or combined use of the FMA module and the LFA module can both achieve enhancements in detection performance.

2) Effectiveness of Dynamic Labels and Gradual Weight: To validate the effectiveness of the LFA module for switching between pseudo-labels and GT labels incrementally with the training process, Table V gives the experimental results for different target-like domain labeling application strategies.

Self-training using only pseudo-labels in the target-like domain is the least effective, with an mAP of only 53.3%. When only pseudo-labels are used, the noise in the pseudo-labels inevitably introduces bias in the learned target detection model. In contrast, when using only GT labels for self-training in the target-like domain, the training effectiveness is improved, with a mAP of 53.9%, but training with GT labels alone does not take advantage of the knowledge synthesized by the teacher model. The absence of knowledge synthesized by the teacher model results in the loss of certain feature information from the target domain. During the training process, combining GT labels with the knowledge synthesized by the teacher model can achieve better results. However, simply mixing GT labels and pseudo-labels did not lead to an improvement in performance. The accurate information in GT labels can be contaminated by the noise present in pseudo-labels, making it challenging for the model to discern correct knowledge from the mixed information. Simply mixing GT labels and pseudo-labels for training fails to fully leverage their potential, making it difficult for the model to adapt to cross-domain foreground-background variations. On the other hand, LFA uses weighted self-training with mixed labels, focusing on GT labels to learn the correct bounding box regression and classification information in the early stages of training, and gradually focusing on pseudo-labels as the training progresses to leverage the teacher's model knowledge, and further improves the mAP to 55.1%.

Using only the LFA module also demonstrates consistent validation performance. When using only pseudo-labels, the mAP is 45.9%, which is the lowest performance. The performance improves when using GT labels, reaching an mAP of 47.4%. Simple mixing of GT labels and pseudo-labels results in only a slight enhancement compared to using pseudo-labels alone. In contrast, the weighted self-training with mixed labels provided by the LFA module achieved the best performance, with mAP increasing to 48.6%.

3) Effect of Hyperparameters: Table II demonstrates the influence of the hyperparameter β in (6) on different domain adaptation tasks. The experimental results indicate that the hyperparameter β has a certain impact on detection performance. When β is set to 0.5, FIE-Net exhibits good performance in domain adaptation tasks across different scenarios. Both increasing and decreasing the value of β result in a decline in the detection performance of FIE-Net. Based on the comprehensive experimental results of the hyperparameter β across various domain adaptation tasks, we have set it to 0.5.

Table III illustrates the impact of the hyperparameters η in (11) on the xView to DOTA-3 adaptive task. We set the weight of $\mathcal{L}_{\text{adv}}$ to 1, while keeping η as the sole analysis parameter. The experimental results of domain adaptation from xView to DOTA-3 indicate that the optimal value of η should be between 0.9 and 1.0. Meanwhile, the experimental results of domain adaptation from UCAS-AOD to CARPK and from HRRSD to SSDD suggest that the optimal value of η should be between 1.0 and 1.1. Taking into comprehensive consideration, the optimal model performance is achieved when the value of η is 1.0.

TABLE VI
ABLATION STUDY OF THE FMA MODULE

Method	Pixel	Instance	Image	mAP_{50}
baseline				45.4
Proposed	✓			47.3
		✓		48.6
			✓	46.9
	✓	✓		50.2
	✓		✓	48.4
		✓	✓	49.7
FMA	✓	✓	✓	**50.9**

4) Effectiveness of the FMA Module: To validate the effectiveness of each component in the FMA module, we conducted ablation experiments by equipping pixel-level, instance-level, and image-level feature alignment subnetworks individually. As shown in Table VI, the introduction of pixel-level, instance-level, and image-level feature alignment subnetworks into the baseline network leads to an improvement in detection accuracy. The pairwise combinations of the three feature alignment subnetworks also enhance detection accuracy, while the application of all three subnetworks to form the FMA module achieved the best accuracy of 50.9%. The ablation experiments on the FMA module demonstrated the effectiveness of its individual components.

E. Comparisons With the State-of-the-Art

In this section, to showcase the effectiveness and generalization capability of the FIE-Net, we evaluate our proposal across three elaborate domain adaptation scenarios: 1) between different optical satellites; 2) between the satellite and the UAV; and 3) cross-modalities from optical to SAR images. In each scenario, we compare the FIE-Net with several state-of-the-art unsupervised domain adaptive methods.

1) Adaptation Between Different Optical Satellites: We first investigated the domain adaptation tasks between different optical satellites. For the adaptation from DIOR to DOTA-10 task, as illustrated in Table VII, the FIE-Net attained the optimal mAP of 57.3%, which is a 16.5% improvement over the baseline and 2.5% higher than the state-of-the-art

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TABLE VII
EXPERIMENTAL RESULTS (%) OF THE SCENARIO BETWEEN DIFFERENT OPTICAL SATELLITES: DIOR → DOTA-10

DIOR → DOTA-10											
Method	Plane	Ship	Storage tank	Baseball field	Tennis court	Basketball court	Ground trackfield	Harbor	Bridge	Vehicle	mAP_{50}
EPM [49]	47.8	46.6	54.7	42.0	85.7	55.5	69.7	23.2	16.6	52.4	49.4
RFA [13]	48.6	49.4	56.8	47.7	86.5	49.3	70.1	19.1	25.3	56.6	50.9
AQT [50]	46.5	41.7	54.1	32.3	85.4	47.0	68.7	11.7	12.4	48.9	44.8
CANet [7]	45.1	38.3	51.3	36.8	84.3	36.9	65.9	11.3	13.9	47.7	42.0
HSA [25]	47.6	43.4	53.9	42.6	85.8	39.2	67.6	13.5	16.3	51.1	46.1
HT [38]	45.8	36.4	58.4	22.3	87.9	20.1	69.1	10.8	6.8	49.9	40.8
SFOD [23]	46.8	39.8	51.1	42.9	83.7	46.5	64.5	15.9	19.3	50.4	46.1
DCLDA [20]	51.3	46.1	58.7	56.6	85.8	**61.5**	68.3	37.3	30.1	52.3	54.8
RST [29]	52.0	46.8	57.6	56.2	86.0	55.5	**71.2**	39.1	27.5	53.6	54.5
Ours	**52.4**	**49.5**	**60.2**	**58.3**	**88.2**	58.9	71.0	**44.3**	**32.2**	**58.4**	**57.3**

TABLE VIII
EXPERIMENTAL RESULTS (%) OF MULTIPLE DOMAIN ADAPTIVE SCENARIOS

xView → DOTA-3					UCAS-AOD → CARPK	CARPK → UCAS-AOD	HRRSD → SSDD
Method	Plane	Ship	Storage-tank	mAP_{50}	mAP_{50}	mAP_{50}	mAP_{50}
EPM [49]	53.5	42.8	31.3	42.5	71.2	65.0	50.9
RFA [13]	57.3	44.7	34.9	45.6	68.4	64.3	49.6
AQT [50]	60.2	49.9	38.9	49.6	71.0	70.8	35.9
CANet [7]	52.1	41.2	32.2	41.8	59.1	59.8	40.2
HSA [25]	62.7	46.5	40.6	49.9	74.8	74.3	58.1
HT [38]	56.8	44.3	35.2	45.4	67.5	67.6	46.8
SFOD [23]	59.9	47.2	38.6	48.6	69.7	70.4	43.5
DCLDA [20]	66.4	50.6	42.1	53.0	76.3	75.7	55.2
RST [29]	**67.7**	48.7	41.9	52.7	75.8	75.2	54.5
Ours	66.8	**52.6**	**45.9**	**55.1**	**79.4**	**78.9**	**58.5**

TABLE IX
EXPERIMENTAL RESULTS OF RUNTIME AND FLOPS

Method	Runtime(day)	FLOPs(G)	FPS(f/s)
EPM [49]	1.6	**187.4**	**13.7**
RFA [13]	1.9	264.3	8.5
AQT [50]	2.1	221.6	11.0
CANet [7]	**1.5**	220.7	11.6
HSA [25]	1.8	253.4	10.7
HT [38]	2.1	232.1	9.4
SFOD [23]	2.2	205.5	12.3
DCLDA [20]	2.0	262.4	9.0
RST [29]	2.6	258.3	7.2
Ours	2.8	261.6	8.1

method. Similarly, as shown in Table VIII, for the adaptation of the xView → DOTA-3 task, FIE-Net achieved the best mAP of 55.1%, which is 9.7% higher than the baseline and 2.1% superior to the state-of-the-art method. This signifies that our algorithm, through the collaboration of FMA and LFA, has made significant strides in the target domain. FMA and LFA emphasize foreground instance features in both feature alignment and label application, assisting the model in achieving improvements in classification accuracy and localization precision.

2) Adaptation Between the Satellite and the UAV: According to Table VIII, it can be observed that the FIE-Net has the highest mAP accuracy of 79.4% in the adaptive task of UCAS-AOD → CARPK, which represents an 11.9% improvement over the baseline and is 3.1% higher than the recent method DCLDA. On the other hand, we have also conducted an adaptation experiment from UAV scenes to satellite scenes on the CARPK → UCAS-AOD task. As shown in Table VIII, our FIE-Net achieves an mAP of 78.9%, which represents an 11.3% improvement over the baseline and is 3.2% higher than the recent method DCLDA. These experiments validate that reducing background interference and emphasizing foreground features are equally effective in domain adaptation tasks between satellites and UAVs. The combination of FMA and LFA effectively overcomes the impact of small foreground-background differences, achieving enhanced domain adaptation detection performance between satellites and UAVs.

3) Adaptation from Optical to SAR Images: We evaluated cross-modalities adaptation from optical to SAR images on the HRRSD → SSDD task. The experimental results are given in Table VIII, where the FIE-Net obtained the best mAP of 58.5%, which represents an 11.7% improvement over the baseline and is 3.3% higher than the recent method DCLDA. In this experiment, we observed that the cross-modalities domain adaptation tasks are characterized by significant interdomain foreground–background discrepancies, posing considerable challenges to cross-domain knowledge transfer. In contrast, our proposal effectively addresses this issue by attenuating the difficulty of cross-domain knowledge transfer. Furthermore, with the combined efforts of FMA and LFA, the model effectively mitigates the adverse effects

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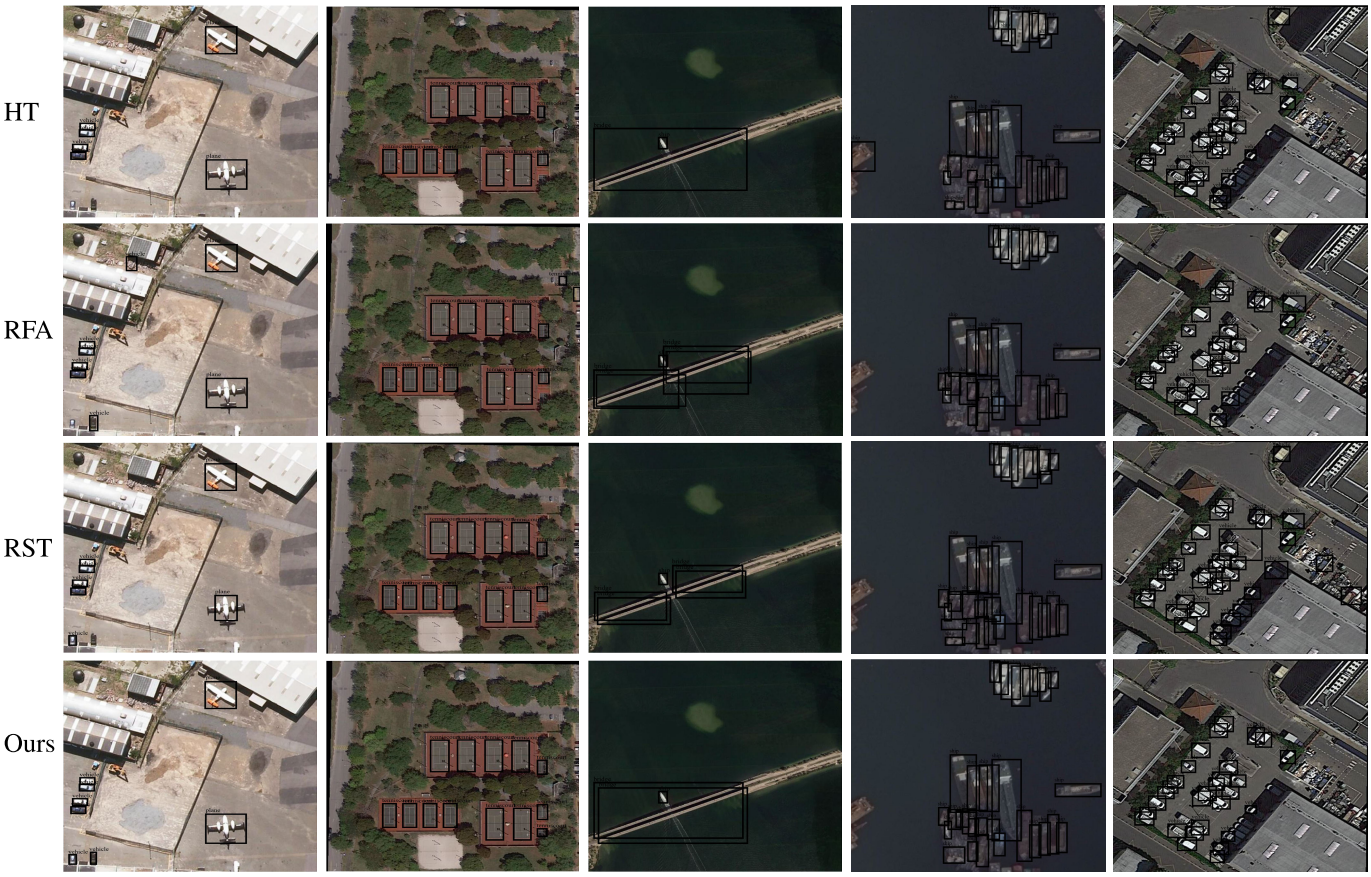


Fig. 5. DIOR → DOTA-10 visualization of adaptive tasks. Our method aligns the domain shift well compared to the baseline method.

caused by large foreground–background discrepancies, achieving an improvement in cross-modalities domain adaptation detection performance. The outcome of this experiment compellingly demonstrates the generalization capability of the FIE-Net.

4) Model Runtime and FLOPs: Table IX presents a comparison of the FIE-Net with other algorithms in terms of runtime and FLOPs. To achieve higher detection accuracy, the FIE-Net involves a more complex model training process. The training time required for the FIE-Net is relatively long, necessitating an additional 4–31 h compared to other methods. The FIE-Net employs an FCOS-based detector in the mean teacher framework, and, following modifications, the model's FLOPs value has increased slightly compared to the baseline network HT. However, the modifications made to the FIE-Net have little impact on inference performance, with the FPS being comparable to that of the baseline network HT.

F. Visualization and Analysis

1) Detection Results: We present the visualization results of the FIE-Net across all domain adaptation scenarios, alongside those of baseline detectors, RFA, and RST. As depicted in Fig. 5, for the adaptation task from DIOR to DOTA-10, the FIE-Net demonstrates more accurate detection results across various categories. For the visualization results presented in the first to third columns, the FIE-Net achieves accurate boundary box delineation. When confronted with mixed targets, as shown in the fourth and fifth columns of the visualization results, the detection performance of the FIE-Net remains relatively unaffected and maintains high accuracy. The FMA module effectively aligns instance-level features, and in combination with the more accurate pseudo-labels generated by the LFA module, they mitigate the impact of biased information in the source domain.

Fig. 6 showcases the visualization outcomes of additional experiments. For the adaptation task from xView to DOTA-3 (columns 1–3), the FIE-Net effectively mitigates false alarms, accurately delineates bounding boxes, and identifies challenging objects that might otherwise be overlooked. In the adaptation tasks between UCAS-AOD and CARPK (columns 4 and 5), the FIE-Net exhibits excellent detection accuracy, highlighting its robustness in simpler domain adaptation detection tasks with minor domain shifts. In the cross-modalities adaptation task from HRRSD to SSDD (column 6), the FIE-Net effectively alleviates the domain shift issue in cross-modalities images. In contrast, other detectors struggle to adapt to the significant domain shift in cross-modalities images, underscoring the powerful generalization capability of the FIE-Net. The intuitive results are in harmony with the numerical evaluations, confirming the superior performance and generalization ability of the FIE-Net across domain adaptation scenarios in remote sensing.

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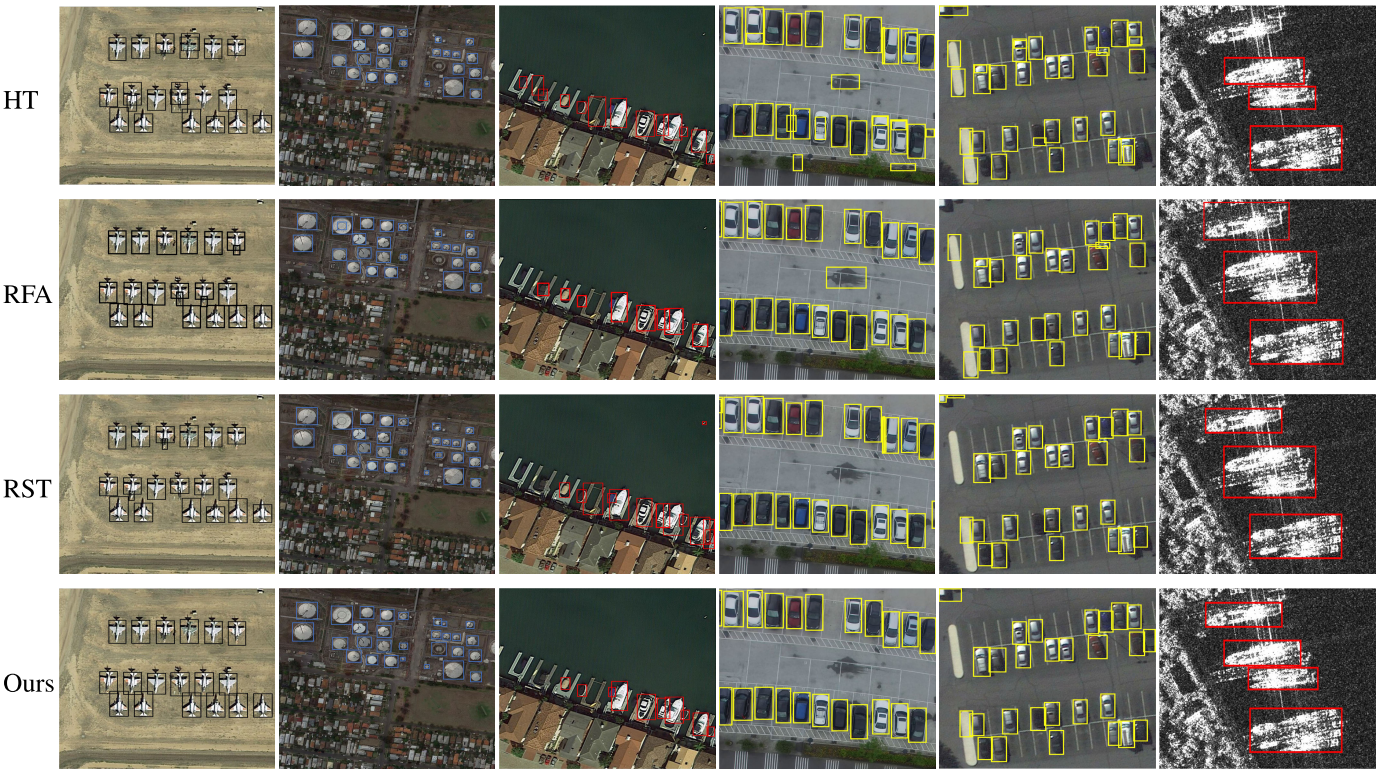


Fig. 6. Comparison of our detection results with state-of-the-art methods in multiple domain-adaptive scenarios. Columns 1–3 represent the adaptation tasks from xView to DOTA-3, while Columns 4 and 5 depict the adaptation tasks from UCAS-AOD to CARPK and from CARPK to UCAS-AOD, respectively. Column 6 shows the cross-modalities adaptation task from HRRSD to SSDD. Different categories are marked with distinct colors: the plane category is in black, the storage tank category is in blue, the vehicle category is in yellow, and the ship category is in red.

When facing domain adaptation tasks between different optical satellites with minimal foreground-background differences, the performance improvement of the FIE-Net is relatively low, with an increase of approximately 2.5% compared to RST. However, for domain adaptation tasks between satellites and UAVs with significant foreground–background differences, as well as for cross-modalities adaptation from optical images to SAR images, the FIE-Net demonstrates a more significant performance improvement, achieving an increase of approximately 4% compared to RST. From Figs. 5 and 6, it can be observed that the RFA performs well in the adaptation tasks from DIOR to DOTA-10 and from xView to DOTA-3. However, it fails to consider the foreground-background differences across different domains, resulting in a high false detection rate in the adaptation tasks between UCAS-AOD and CARPK. Additionally, the RFA makes it difficult to divide the bounding boxes correctly in the cross-modalities adaptation task from HRRSD to SSDD.

The FIE-Net demonstrates improvement in the detection performance of isolated objects. Additionally, it also exhibits strong detection capabilities for partially occluded and overlapping targets. From the lower half of Fig. 5, it can be observed that FIE-Net shows limited optimization in detecting small targets that are densely distributed within a confined area. We believe that the presence of densely packed small targets affects the FMA module, leading to the modules failing to achieve their intended effects.

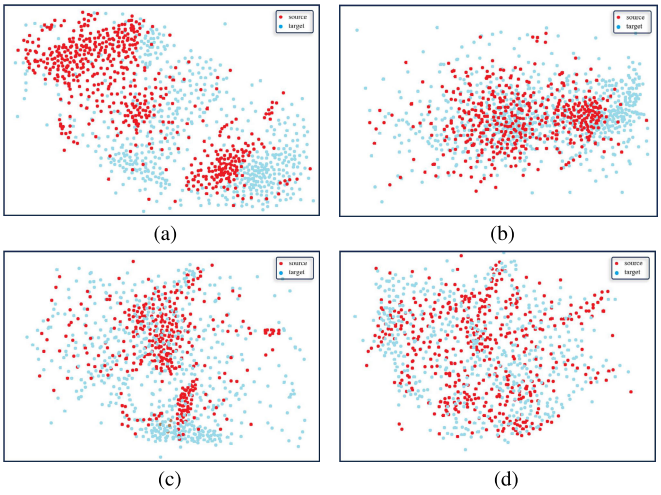


Fig. 7. Feature visualizations. Our method aligns the domain shift well compared to the other method. (a) Baseline detector. (b) RFA. (c) RST. (d) FIE-Net.

2) Feature Visualization: Using the t-distributed stochastic neighbor embedding (t-SNE) method [51], we visualized the extracted backbone features on the xView → DOTA task, as shown in Fig. 7. Obviously, baseline detectors are difficult to get rid of the interference from the background, have significant cross-domain gaps, and are easy to distinguish different domain features. In contrast, our proposed FIE-Net effectively mixes different domains, enabling the detector to

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extract instance domain invariant features and improve cross-domain performance.

IV. Conclusion

This article introduces the FIE-Net, an unsupervised domain-adaptive object detector designed for remote sensing scenarios. Our proposed FIE-Net is based on the mean teacher framework and consists of two parts: the foreground-focused multigranularity feature alignment (FMA) module and the LFA module. The FMA module performs feature alignment at three granularities: pixel-level, instance-level, and image-level. Piggybacking the foreground focusing module on the pixel-level and instance-level feature alignment reduces the background-induced negative migration, to highlight the foreground instance features. In addition, the LFA module contains a hybrid threshold for the label filtering module and a progressive label-switching strategy. The hybrid threshold for the label filtering module can balance the foreground–background differences in the information contained within the labels. The progressive label-switching strategy makes the process of transferring knowledge from teacher mode to student mode more effective. Experiments conducted in remote sensing domain adaptive scenarios show that FIE-Net achieves superior performance, validating the effectiveness and generalization ability of the method. We hope that our work can inspire further research in the domain of adaptive object detection for remote sensing images.

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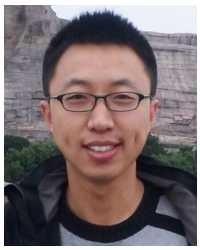
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